

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	5 MARKS

4.2 – Proposed Solution

The proposed solution focuses on leveraging a modern data analytics ecosystem and Tableau's advanced visualization capabilities to address the challenges associated with unhealthy dietary habits among college students. The solution begins with the collection of data from multiple sources, including student surveys, cafeteria transaction records, nutritional databases, mobile food-ordering applications, and demographic information. These diverse datasets are integrated into a centralized database system using data engineering and ETL (Extract, Transform, Load) processes to ensure accuracy, consistency, and reliability. Once the data is prepared, Tableau is utilized to create interactive dashboards that provide real-time insights into students' food preferences, meal consumption patterns, calorie intake, nutritional balance, and spending behaviour. The dashboards enable stakeholders to analyze trends across different student groups, such as on-campus residents, off-campus students, and commuters, helping identify key factors influencing food choices. The analytical framework supports evidence-based decision-making by highlighting nutritional deficiencies, unhealthy eating patterns, and opportunities for intervention. Based on these insights, the proposed solution recommends implementing personalized dietary strategies, including customized meal recommendations, healthy eating awareness campaigns, budget-friendly nutritious meal plans, and improvements to campus dining services. Additionally, predictive analytics can be incorporated to forecast future dietary trends and identify students who may be at risk of poor nutritional outcomes. Regular monitoring through Tableau dashboards allows institutions to measure the effectiveness of

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dietary programs and continuously refine their strategies. By combining data integration, business intelligence, and visualization technologies, the proposed solution transforms raw dietary data into actionable insights that promote healthier eating behaviors, improve student wellness, and support academic success. Ultimately, this comprehensive approach creates a sustainable, data-driven framework that empowers colleges and universities to foster a healthier campus environment while optimizing food service operations and enhancing the overall student experience.

Proposed Solution: Comprehensive Analysis and Dietary Strategies with Tableau

A College Food Choices Case Study

A data-driven solution to understand college students' food choices, nutritional intake, and spending behavior using a modern data stack and Tableau dashboards to design effective dietary strategies.

